

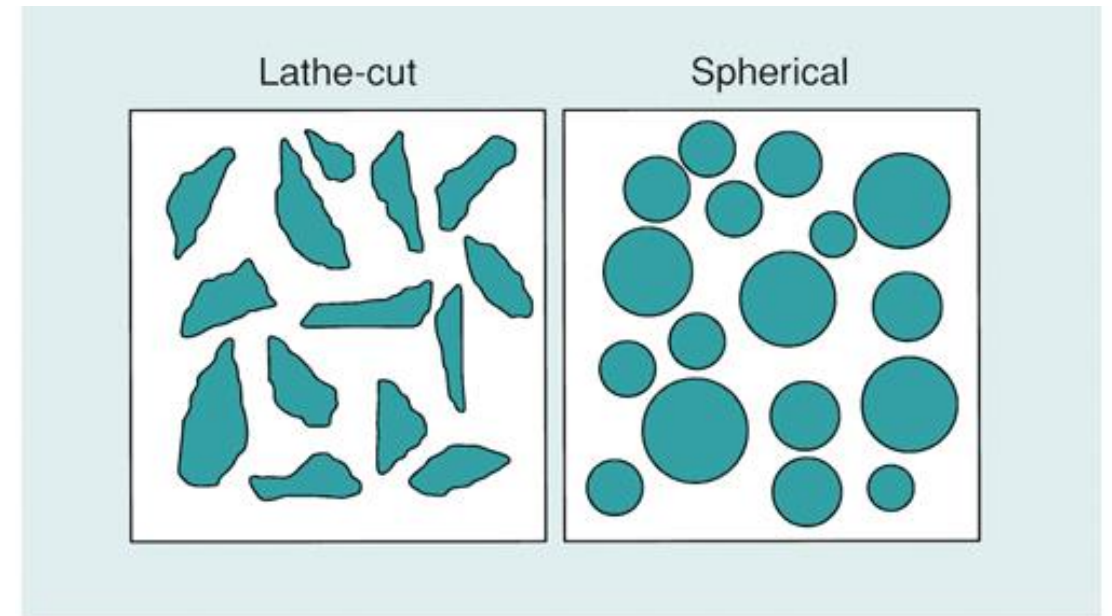


Amalgam and Resin-based Composites

Jayden Yoon, DDS

What is amalgam?

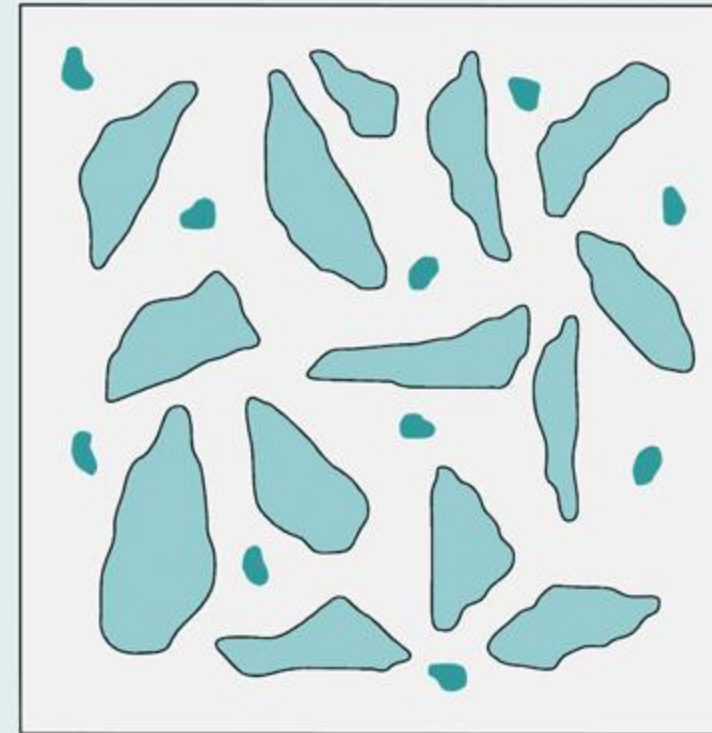
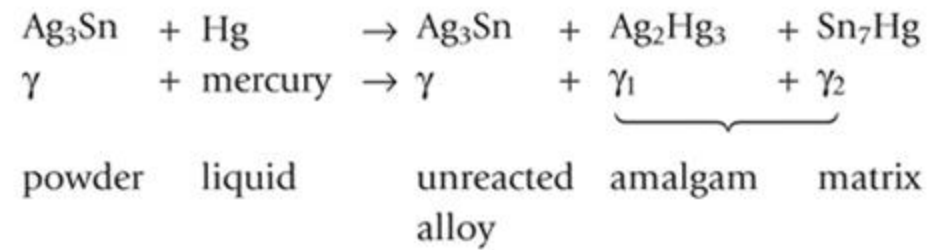
- The hard, rigid product formed when Hg reacts with an alloy comprised of Ag, Sn, Cu
- Comes in capsules separating liquid Hg with alloy powder
- Particles can come in various shapes and sizes
- Powder particles can come in lathe-cut, spherical, or admixed/dispersed-phase alloy (combination)



Amalgam corrosion and composition

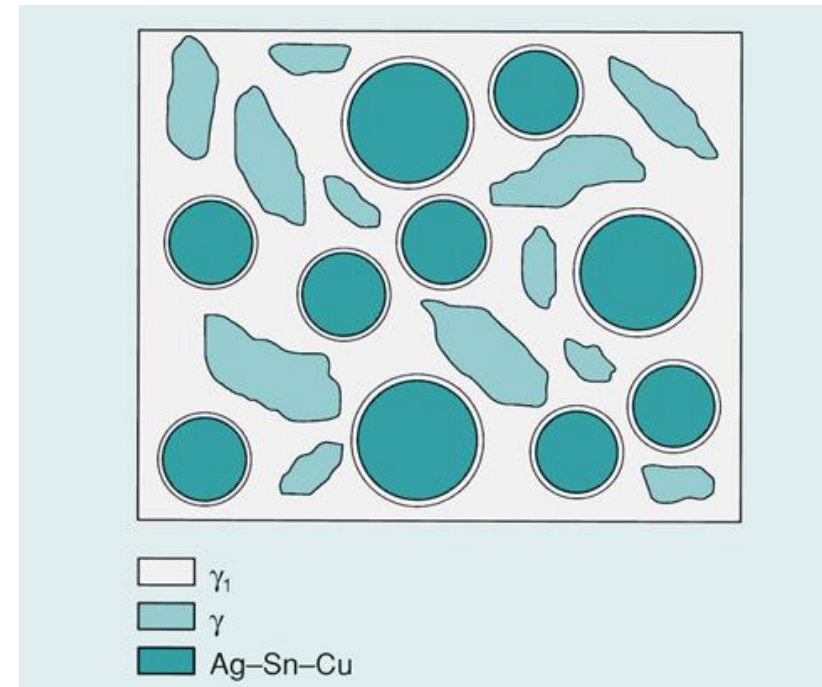
- Typically known to corrode over time
 - Excessive corrosion leads to restoration failure (already covered)
 - Minimal corrosion allows byproducts to build up in microscopic voids between restoration and tooth, sealing the interface (crevice corrosion)
- 40-50% Hg, 50-60% alloy
 - Ag & Sn alloy
 - Cu increases strength and hardness, reduces corrosion

Amalgam reaction



Copper added to modify properties

- Gamma-2 generally has undesirable characteristics
- Copper reduces Gamma-2 by converting some to Gamma-1



What makes an amalgam stronger?

- Want the most gamma phase possible (unreacted with Hg)
- Less gamma-1 and gamma-2 allows more compressive and tensile strength
 - Copper helps reduce gamma-2, also reducing effects of creep and corrosion
 - Amount of gamma-1 is dependent on amount of Hg in capsule to

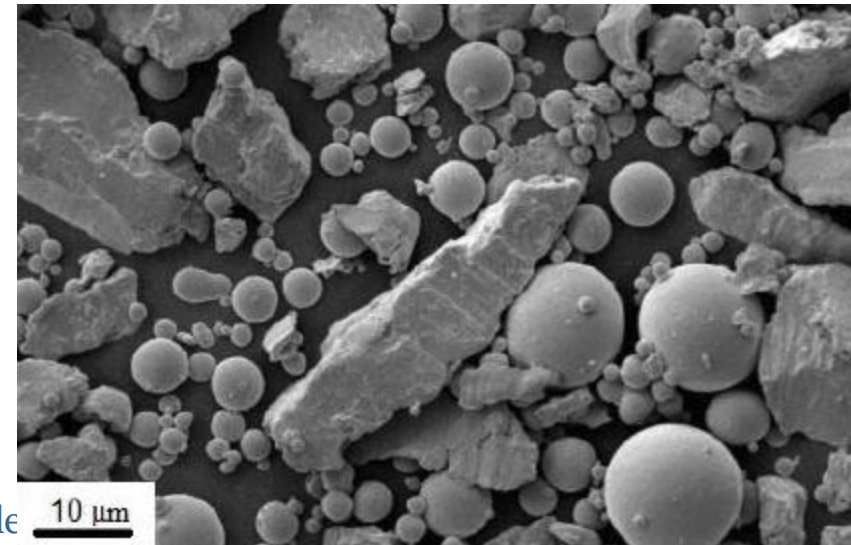
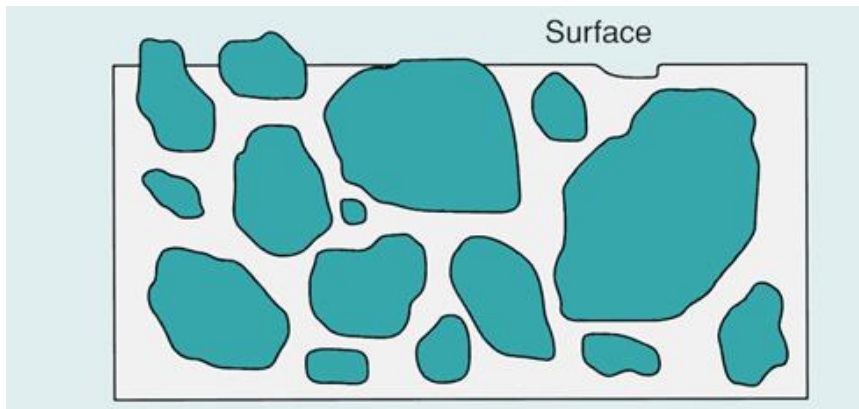
TABLE 2.1.3

Tensile Strengths of Phases of Amalgam

Phase	Tensile Strength (MPa)
γ	170
γ_1	30
γ_2	20
Amalgam	60

Selection of amalgam properties

- Particle size
 - Small: easily manipulated, but reacts with Hg more (more gamma-1 & gamma-2) and thus requires more Hg in the capsule
 - Large: more strength, but not as easily manipulated
 - The larger the particle, the more easily it can be dislodged when a dentist carves
 - Best: medium / admixed



Properties

- High thermal conductivity
- Galvanic effects - when amalgam contacts another metal (i.e. gold) in a conducting medium (saliva), the more electronegative metal breaks down --> metallic taste in mouth
- No bonding – retention only (undercut preps)
- Need large preparations to strengthen the amalgam (to withstand tensile forces) --> weakened tooth structure



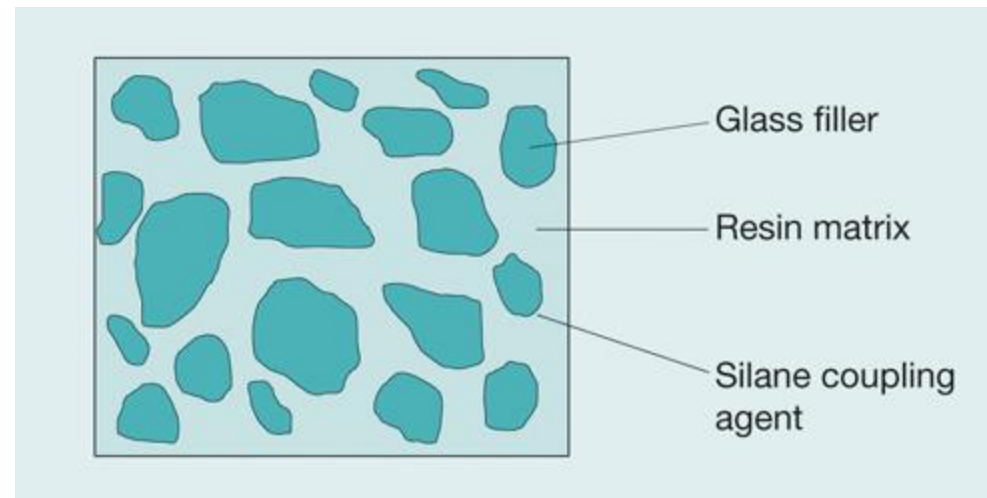
Amalgam and health

- Hg in pure form is considered a hazard, but when bound and reacted in the amalgam, it is considered safe and below hazardous threshold
 - Technically, there is minimal Hg exposure during placement and removal of amalgam
- Small changes in the amalgam – breakdown, corrosion, etc. - make way for minimal exposure



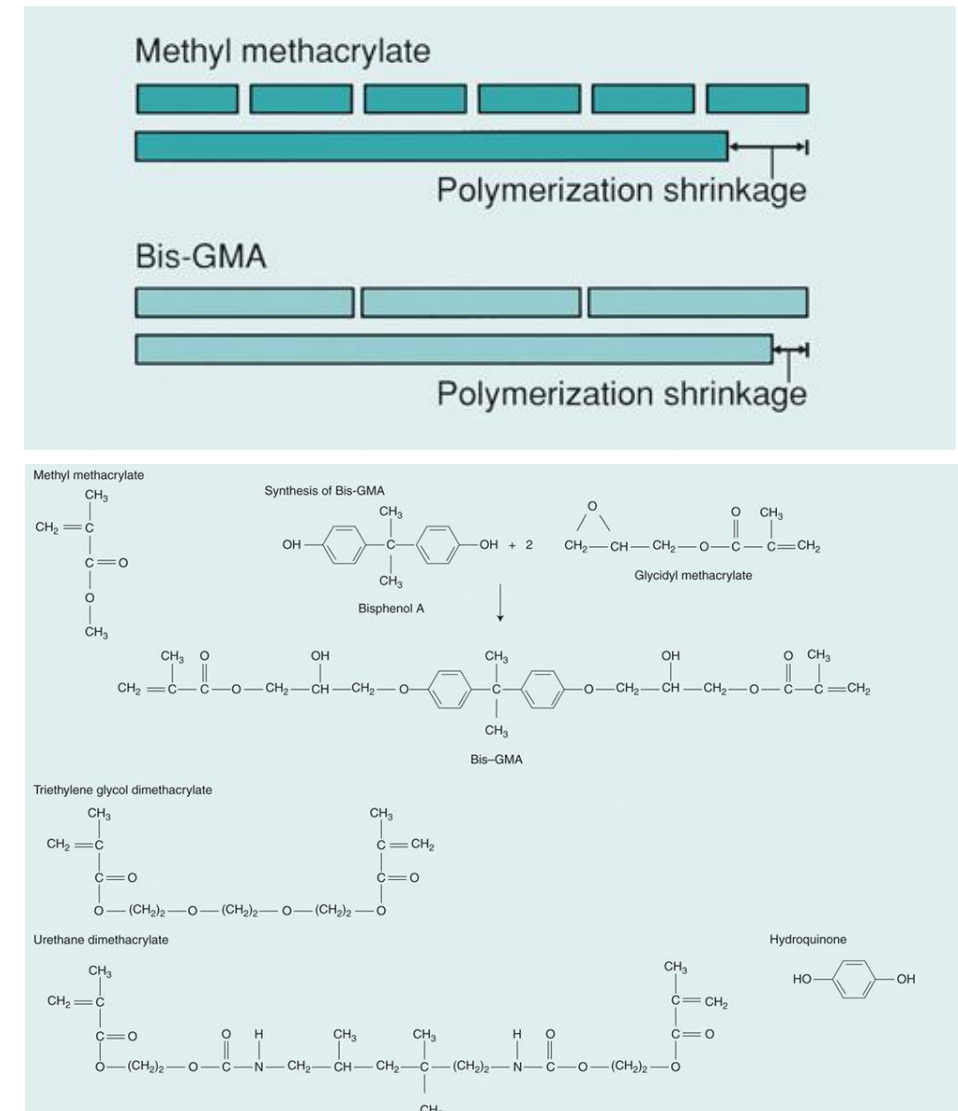
Resin-based composites

- Components: organic resin, inorganic filler (most of composite's mass), coupling agent



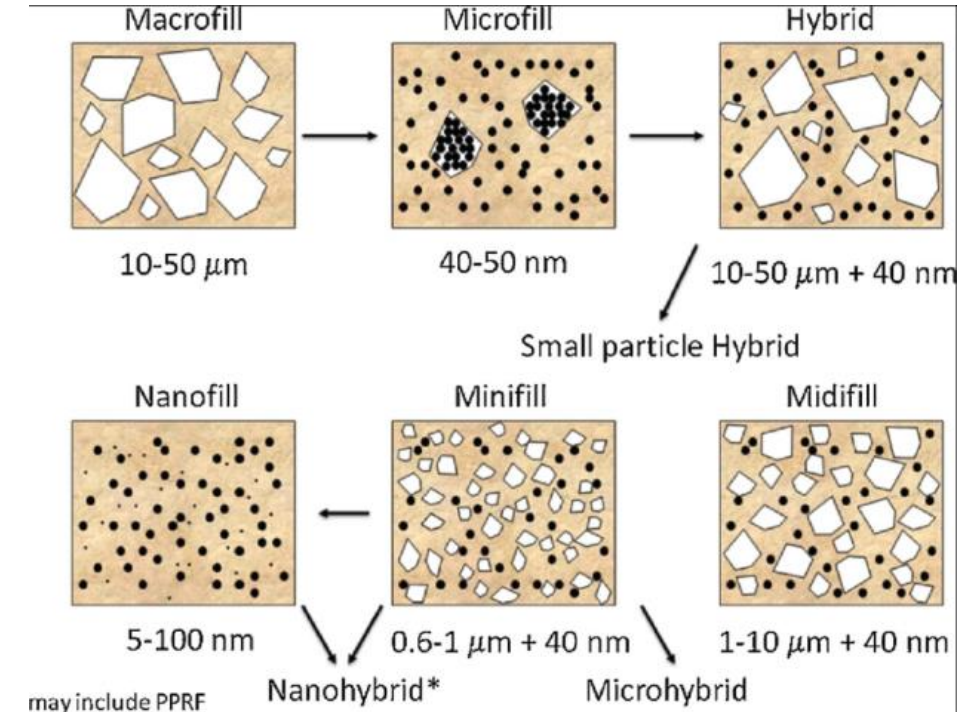
Resin matrix

- Fluid monomer --> solid polymer through radical-mediated addition polymerization
- Common polymers in composites are UDMA (urethane dimethacrylate) or bis-GMA (bisphenol A-glycidyl methacrylate)
 - Large MWs mean more viscosity, less shrinkage
 - Use TEDGMA (triethylene glycol dimethacrylate) as a viscosity control
- Hydroquinone (inhibitor) present to prevent premature polymerization



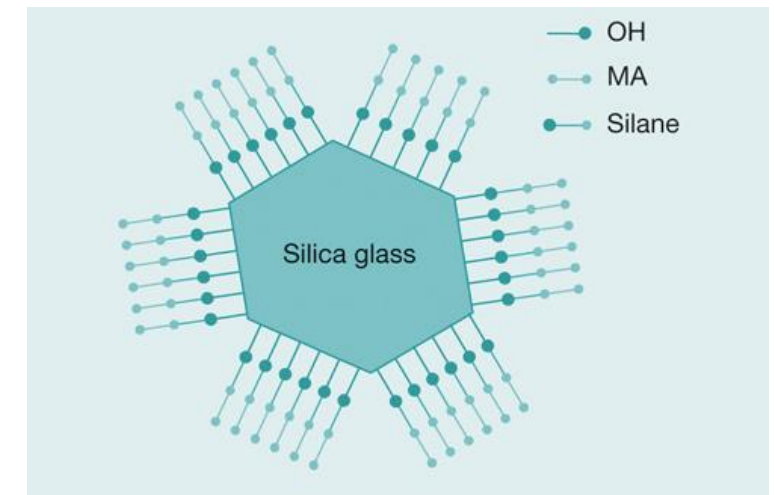
Filler

- Helps reduce shrinkage – filler does not partake in polymerization of resin, and introducing filler reduces overall proportion of resin
- Coefficient of thermal expansion close to teeth - more filler leads to a coefficient closer to that of teeth
- Mechanical benefits: hardness, compressive strength
- Salts of heavy metals (barium, strontium) provide radiopacity
- Esthetic control of color, translucency, and fluorescence
- Usually composed of quartz or silica-based glass
- Smaller filler particles --> smoother finish
- Materials: glass/quartz, colloidal silica, zirconia-silica



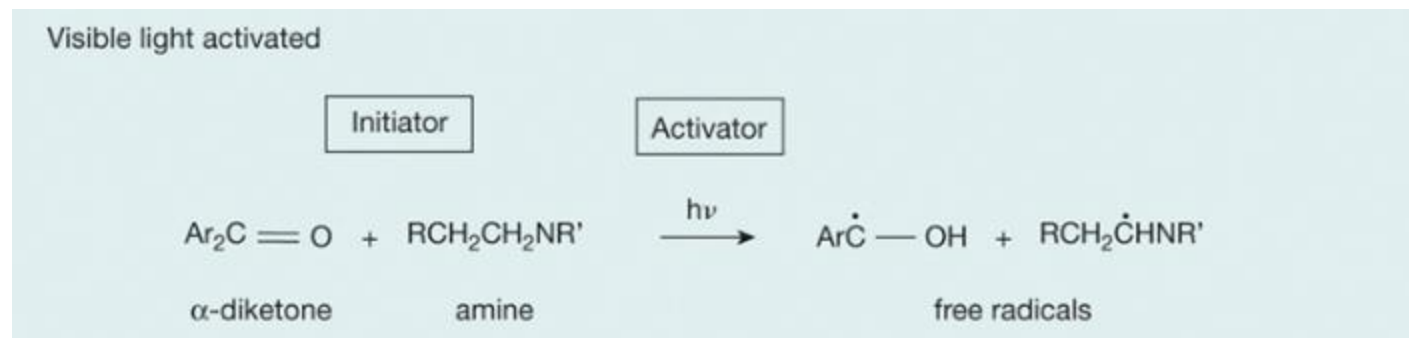
Coupling agent

- Silanes that create strong bond between filler and resin
 - If bond is weak or nonexistent, only the resin takes loading forces and filler-resin interfaces are crack initiation sites -- > fracture and wear
- Most common is gamma-MPTS (methacryloxypropyltriethoxysilane)



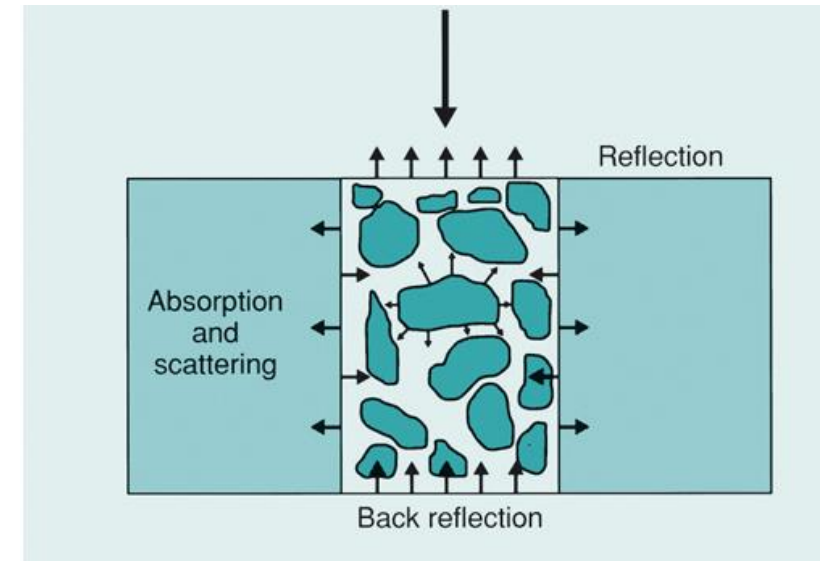
Resin properties

- Includes camphorquinone (initiator); when exposed to visible light (activator) of 460-480 nm addition polymerization occurs
- After curing, resin-oxygen interface creates oxygen-inhibited layer (uncured resin)
 - OK when layering composite during restoration – allows further polymerization and allows inclusive polymerization of new composite
 - Needs to be removed (wiped, polished) at end of restoration



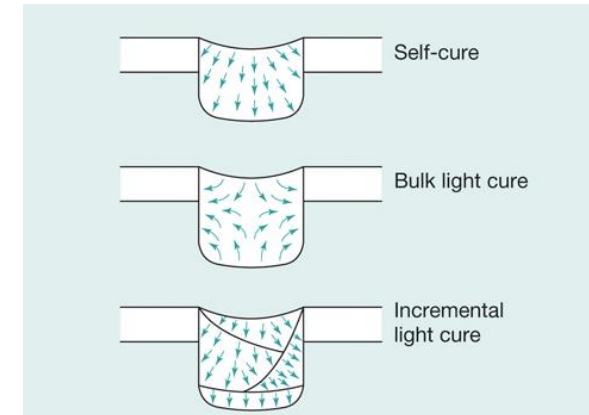
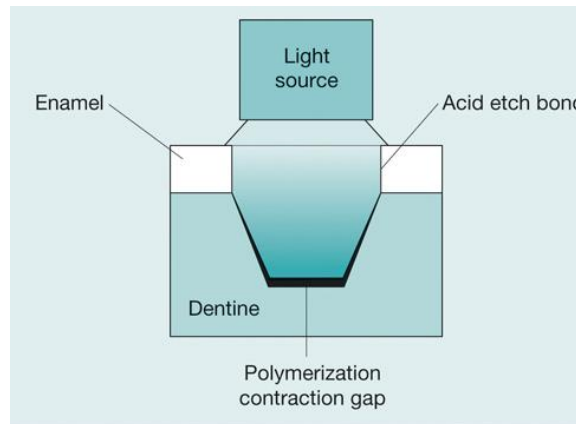
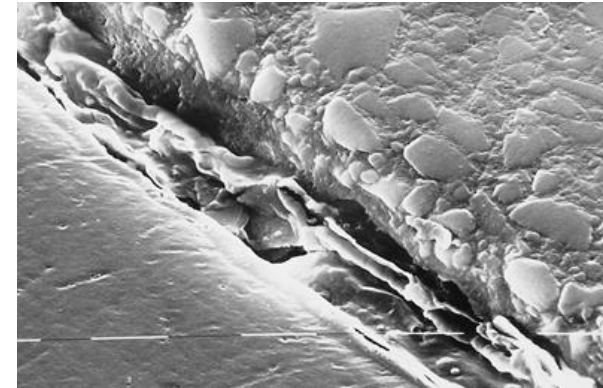
Properties that affect curing

- Type of composite
 - Darker shades absorb more light - less curing depth
 - Filler shape and size - more reflected light means less light penetration
- Quality of light source
- Distance between light and composite
 - Have curing light as close as possible



Polymerization shrinkage

- Shrinkage can never be fully eliminated
 - Most apparent with dentin-resin interface
 - Post-op sensitivity due to hydrodynamic effect
 - Marginal leakage if restoration margin is on dentin
- Reducing shrinkage
 - More filler: less resin in the material
 - Smaller increments



Flowable and packable composites

- Flowable – low viscosity, reduces voids in small areas
 - Lower viscosity by incorporating less filler proportionately, generally made of larger filler particles
 - Therefore, cannot withstand a lot of stress and wear
- Packable – higher viscosity, easier to manipulate
 - Greater viscosity and properties by incorporating
 - Filler that may be larger, shaped to allow interlocking
 - Resin with stronger intermolecular attraction
 - Dispersants (additive that lowers viscosity), allowing more filler addition to balance
 - More filler = poorer esthetics (opaque), but stronger



Bulk fill composites

- Curing depth 4-8mm possible
- Created with a fine balancing act of:
 - Reducing filler amount, allowing more light penetration
 - BUT: increasing filler size to not compromise strength
 - Incorporating other photoinitiators that may activate polymerization with longer (weaker) wavelengths



Classification based on filler

- Filler size and finish
 - Larger filler: less filler surface area overall, thus less resin needed in composite; poorer finish
 - Smaller filler: more filler surface area, thus more resin needed to cover the filler; smoother finish
- Goal: as much filler loading as possible without compromising finish

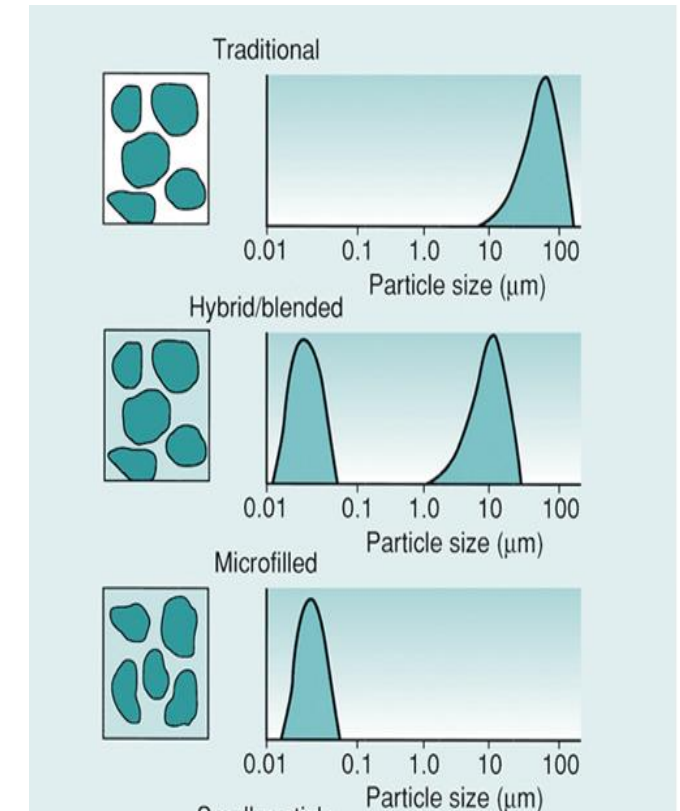
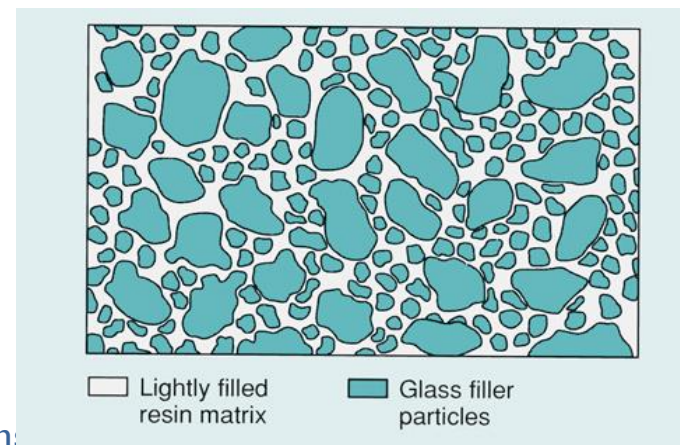
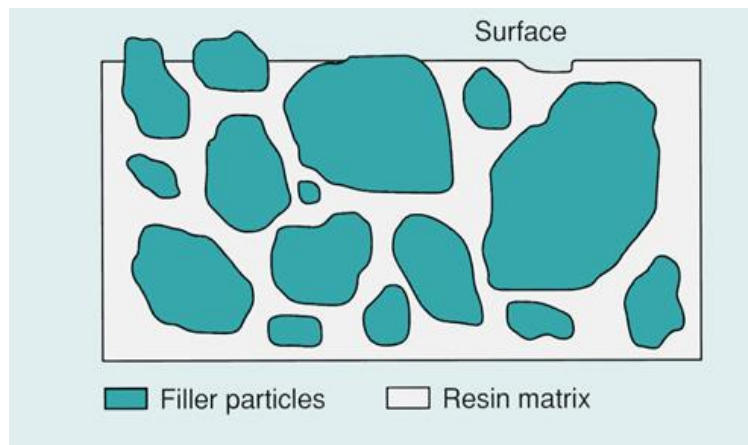


TABLE 9.2 Characteristics of Various Types of Resin Composites

Type of Composite	Size of Filler Particles (mm)	Volume of Inorganic Filler (%)	Handling Characteristics and Properties	
			Advantages	Disadvantages
Multipurpose	0.04, 0.2–3.0	60–70	High strength, high modulus	
Nanocomposite	0.002–0.075	72–79	High polish, high strength, high modulus, polish retention	
Microfilled	0.04	32–50	Best polish, best esthetics	Higher shrinkage, lower strength
Bulk filled	0.04, 0.2–20	59–80	Deep cure, reduced step	Questionable marginal adaptation
Flowable	0.04, 0.2–3.0	42–62	Syringeable, lower modulus	Higher wear
Laboratory	0.04, 0.2–3.0	60–70	Best anatomy and contacts, lower wear	Laboratory cost, special equipment, requires resin cement

Craig's Restorative Materials

General properties

- **Biocompatibility**
 - Better polymerization (proper curing) prevents loss of byproducts into oral environment
 - Generally, amount of byproducts released is extremely low
 - Lower MW molecules (MMA, TEGDMA) have a slightly higher chance of leaching out, compared to higher MW molecules like bis-GMA and UDMA
- **Water sorption**
 - Can lead to issues with color stability and wear resistance, especially if voids are in restoration, improper isolation
- **Coefficient of thermal expansion**
 - Generally close to that of tooth structure
 - More filler contributes to lower coefficient of thermal expansion
- **Radiopacity**
- **Color match** - discoloration from marginal gaps, water sorption, and roughness



Mechanical properties

- Compressive strength
 - Direct restoration materials are stronger than tooth tissue
- Aspect of failure likely tensile strength, since restorations are brittle
- Fair hardness – not likely to experience localized deformations
- Experiences wear
 - Abrasive - when 2 or 3 bodies abrade each other, softer surface is damaged
 - Fatigue / Creep - cracks after repeated loading
 - Corrosive - acidic attack, hydrolysis, breakdown of resin (and resin-filler interface)

TABLE 2.2.3

Compressive Strength Data for a Variety of Materials

Material (Manufacturer)	Compressive Strength (MPa)
Molar enamel	260
Molar dentine	305
Sybralloy (Kerr)	500
Dispersalloy (Johnson & Johnson)	440
G-aenial Posterior	323
Tetric Evo Ceram	282
Filtek Bulk Fill	665
Surefil SDR Bulk Fill	1019



Dental Lab Composites

- Indirect composite (veneers, inlay, onlay)
 - Composites fabricated in the lab, then delivered by dentist
 - Helps account for ideal morphology, curing, and polymerization shrinkage
- Fiber-reinforced composites
 - Better flexural strength, impact resistance
- Composite-ceramic hybrids



Clinical considerations

- Replacing amalgam restorations (larger preparations by nature) - be mindful of layering
- Lack of enamel in large defects - likely marginal gap formation
- Moisture control - cannot bond
- Bruxing - high wear



Compomers

- Newer resin-composite product that releases fluoride
 - Carboxyl groups in "special resin" perform acid-base reaction with the glass, releasing fluoride
 - Hydrophilic components help transport the fluoride using water from mouth
- Hybrid of composites and GIC

TABLE 2.2.5

Composition of a Compomer

Component	Function
Fluoro-alumino-silicate glass	Filler and a source of fluoride
Dimethacrylate monomer (e.g. UDMA)	Forms the resin matrix
Special resin	Provides carboxyl groups
Hydrophilic monomers	Aid the transport of water and fluoride
Photoactivators/photoinitiators	Provide cure by radical polymerization